

On the use of Force Sensitive Resistor Sensors Towards the Improvement of Comfort in Canine Limb Prostheses

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Abstract—Canine limb amputations cause, in many cases, the use of prostheses to restore limb function and improve the quality of life. However, prosthesis use may also cause discomfort problems due to the poor distribution of forces in the socket of the prosthesis, which causes skin lesions, pain, infections, and other complications. Therefore, a system for monitoring the forces between the socket and the residual limb is critical to extract information to improve socket and prosthesis design. This paper presents a portable force monitoring system using Interlink Electronics FSR 402. The system has a user interface that shows, in real-time, in a 3D model, the forces applied on the residual limb through a heat map. In addition, the user interface features force graphs over time, allowing analysis over long periods. Results indicated that implementing this system would help improve socket and prosthesis designs of amputated canine limbs.

keywords—Force Sensing Resistor (FSR), pressure measurement, prosthesis, socket design, residual limb.

I. INTRODUCTION

According to the World Health Organization, in developing countries, there are around 40 million amputees, and 5% have access to prosthetic care [1]. Amputations produce significant physical disabilities in many people and also animals. Even though quadrupedal animals like canines can walk with three of their legs, this could also generate other complications and decrease the quality of life [2].

Vets suggest amputations as recommended methods for treatments of neoplasm, traumatic injuries, peripheral neuropathies, infections, disabilities derived from degenerative processes, paralysis, arthritis, congenital deformity. Related to canines, amputations are produced mainly for traumatic injuries (65%) and neoplasm (35%) [2], [3], [4]. Moreover, osteosarcoma is one of the most common tumors in canines. According to the National Canine Cancer Foundation USA, it is estimated that there are 6000 to 8000 new cases per year [5], with amputation being one of the most recommended methods for these types of tumors.

Over time, limb amputation in canines and the use of prostheses are increasing [4]. Amputees often develop some issues associated with the prosthesis socked, such as discomfort, pain, skin irritation, infections, ulcers, among others [6], [4]. With regards to humans, skin lesions occur in 60% of lower limb amputees, which induces that around 25% of amputees reduce the use of prostheses [7]. These types of complications are mainly produced by poor distribution of forces.

For the prosthesis design, it is crucial to consider many aspects such as the shape of the socket, materials, alignment, suspension system. These parameters determine the stress and strain over the contact interactions between the socket and the residual limb [8], [4]. Hence, it is necessary to understand the interaction between the stump and the prosthesis socket employing a monitoring force system distributed over the stump. Thus, it is possible to modify the design parameters of the socket to obtain a more comfortable and effective prosthesis.

Although some commercial measurement systems are specially designed for pressure mapping in prosthetics, such as the Tekscan F-socket [9] that uses proprietary thin-film sensors, their cost may make them unaffordable for animal applications. Some studies attempt to create a pressure monitoring system as shown in [10], [11]; however, those systems are not portable, limiting their use to some laboratory environments only. Other works proposed wireless solutions [7], [12]. Nevertheless, the systems do not possess a 3D model displaying the data in real-time, which would allow better visualization and interpretation of the data.

Therefore, this work aims to develop an affordable portable system based on low-cost commercial pressure sensors for monitoring forces between the socket and the stump of the amputated limb to improve the comfort and limb function of the user, in this case, a canine. A user interface is developed to monitor the forces applied over a 3D model of the dog stump in real-time, using a heat map. Moreover, graphs of force vs. time are also presented to help analyze the data and improve the mechanical design. In addition, the portability of the system allows performing more complex tests in different scenarios.

This paper is organized in the following manner: section II presents a description of the system, including sensors, conditioning circuits, power supply, data acquisition, and WiFi communication and user interface. In section III, the results of real-time monitoring forces are described. The overall findings are presented in the conclusions section.

II. METHODOLOGY

In this section, prosthesis design is explained as well as each part of the electronic design to collect force data in real-time over a stump and monitor it through an interface in LabView.

A. Design of the prosthesis

The most critical component of a prosthesis is the prosthetic socket. Similar to human prostheses, socket design in the canine limb-prosthesis interface should properly achieve satisfactory load transmission, stability, comfort, and efficient control for mobility [4]. In this study, a custom socket for a canine with an amputated posterior left limb was built.

The exo-prosthesis was designed and developed utilizing a 3D scanner to capture limb topography from a cast of the residual limb for digital sculpting and computer modeling [4], [13]. A negative cast was first obtained from the stump of the patient using alginate mass in order to obtain a positive cast made of plaster following the process described by Machoczek (2021) [14]. The plaster cast was then 3D scanned from a 360-degree view using a Zirkon Zahn Scanner to design a customized prosthesis based on a 3D computational model, as shown in Fig. 1. The prosthetic socket was built using additive manufacturing on a 3D printer using polylactic acid (PLA) plastic. In addition, a prosthetic liner was built from a thin silicon layer covered with fabric as an interface between the socket and the stump to maximize comfort. Finally, a system of elastic straps that extend towards a chest harness was implemented to hold the prosthesis in place, as shown in Fig. 2.

Although functional and empirical results indicated that the custom-designed, 3D-printed prosthetic parts, which included the pylon (foot and the body) of the prosthetic, were strong enough to withstand the high-impact forces and stresses experienced during the canine gait. Knowledge of the interactions among the socket, liner, and the underlying stump remained unclear. Therefore, a network of force sensors was embedded in the socket to understand such interactions better and improve the prosthesis design.

B. Electronic design

As is presented in Fig. 3, the system consists of four components: (i) a network of sensors for monitoring forces, (ii) a multiplexer and conditioning circuit, (iii) an ESP 32S as micro-controller in charge of collecting, processing and communicating data, and (iv) a graphical interface.

1) *Sensor*: The sensor used in this prototype is the FSR (Force Sensing Resistor) 402 of Interlink Electronics [15]. The active area of the sensor is 1.27 mm^2 which corresponds to a diameter of 12.7 mm. This sensor reduces the resistance while the applied force over the active area increases. The FSR is optimized for touch control applications of electronic devices, medical systems, and robotics [15]. The sensors have been used in various biomechanical applications of prosthetic control and pressure measurements. However, their limitations in terms of precision and repeatability must be taken into account [16].

Some measurements were performed for sensor calibration. Distinct forces were applied over three sensors, and the resistance response of each one was collected. According to the data-sheet [15], $10 \text{ M}\Omega$ is the resistive response when there is no force applied over the sensor. Fig. 4 presents the

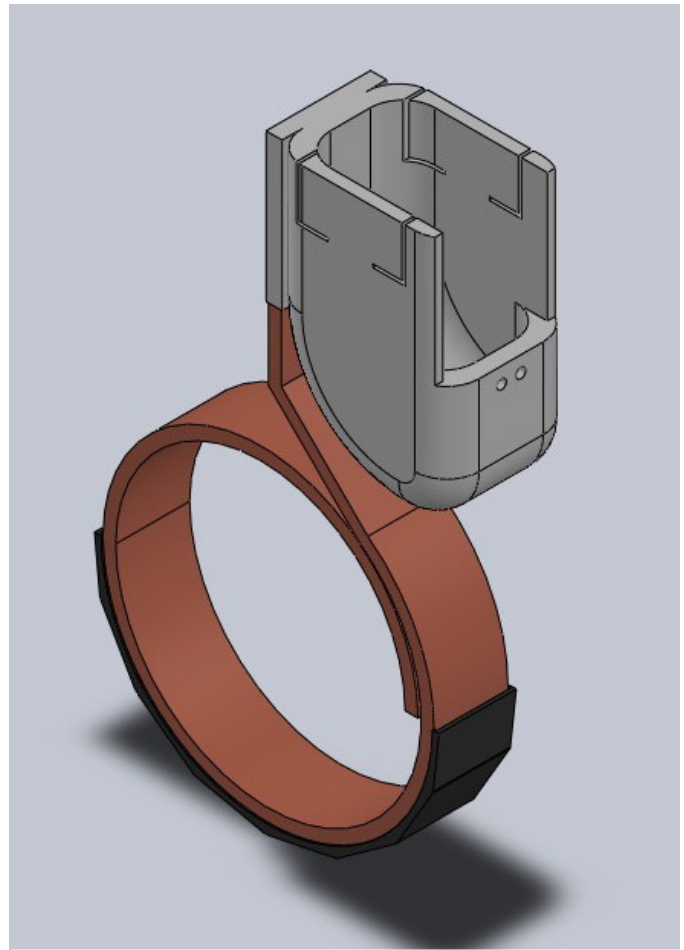


Fig. 1: 3D Model of the Prosthesis.



Fig. 2: Canine prosthesis fitted on patient.

resistance measurements for distinct forces between 0.7 N to 15 N.

2) *Conditioning circuit*: A transimpedance amplifier was selected as a conditioning circuit. This configuration allows to convert current to voltage and amplify the output working with the conductance instead of resistance. Conductance is directly proportional to applied force [17]. In addition, this configuration is suggested by the sensor manufacturer because

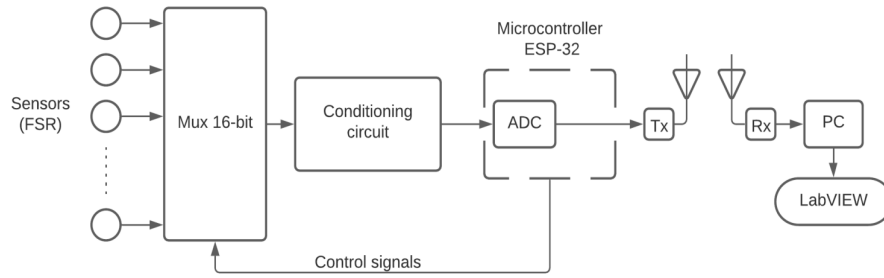


Fig. 3: System description diagram.

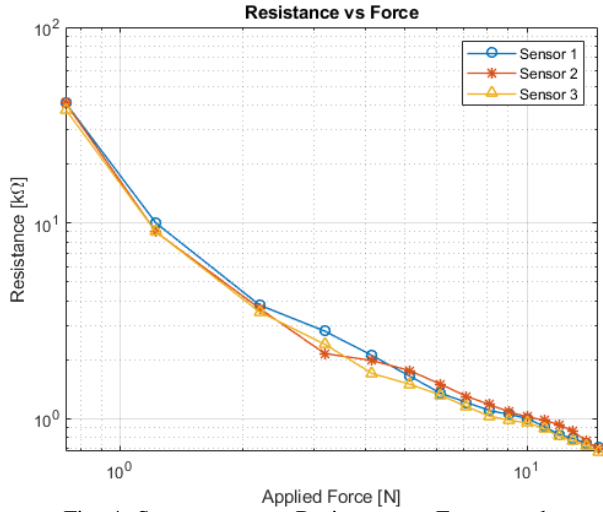


Fig. 4: Sensors output: Resistance vs Force graph.

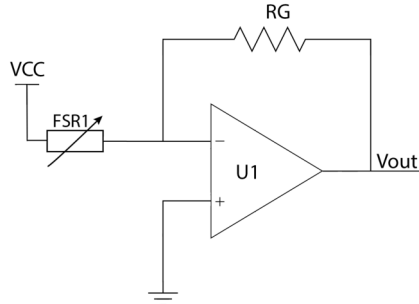


Fig. 5: Circuit used for the transimpedance amplifier.

the output voltage maintains a linear relationship with the applied force. Fig. 5 shows the suggested circuit.

The output voltage of the circuit is described by (1), where R_G is the feedback resistance, and R_{FSR} is the resistive response of the sensor.

$$V_{out} = V_{ref} * -\frac{R_G}{R_{FSR}} \quad (1)$$

However, as the system is designed to be portable by using a battery, some changes were made. The final circuit can be observed in Fig. 6 which presents the conditioning circuit formed by the FSR sensors, followed by the multiplexer HP4067 and the transimpedance amplifier.

This final configuration works with only one voltage source. V_{SS} is the battery voltage (7.4 V), while V_{CC} (5 V) is the voltage for the conditioning circuit and multiplexer obtained

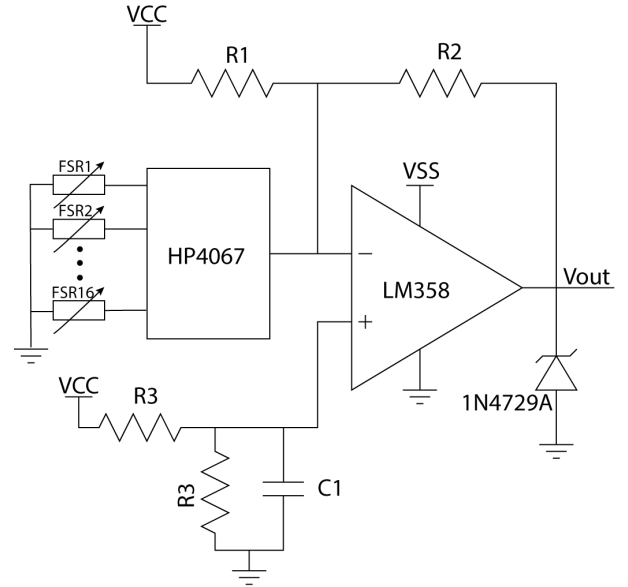


Fig. 6: Conditioning circuit

from the same battery using a boost converted step-down circuit. Thus, the conductance is presented by (2) and voltage output by (3).

$$G_s = \frac{1}{R_{FSR}} = \frac{2}{R_2 * V_{CC}} \left[V_{out} - \frac{V_{CC}}{2} \left(1 - \frac{R_2}{R_1} \right) \right] \quad (2)$$

$$V_{out} = \frac{R_2 * V_{CC}}{2} \left[\frac{1}{R_{FSR}} + \frac{1}{R_2} - \frac{1}{R_1} \right] \quad (3)$$

The response of this circuit is linear so that the voltage is proportional to the applied force. R_2 is the resistance that changes the slope of the voltage vs. force curve, while R_1 aids in establishing the offset of the output voltage. The resistance data is analyzed for the choice of R_1 and R_2 values. The force working range for this application was selected from 0 to 11 N. Therefore, the final values are $R_1=1.3 \text{ K}\Omega$ and $R_2=1.2 \text{ K}\Omega$.

The integrated circuit LM358 is the selected operational amplifier as recommended by the FSR sensor datasheet because it has a good performance with this type of sensor. Also, it needs only one polarization source, which is advantageous in this project due to its portability. The performance with only one voltage source requires a virtual ground, which is obtained by dividing V_{CC} by two [18] using a voltage divider. The values

for resistance selected were: $R_3 = 100 \text{ k}\Omega$ and a capacitor for reducing the noise of $C = 0.1 \text{ }\mu\text{F}$.

The final system has 16 sensors, and thanks to the use of a multiplexer, it is possible to have only one analog to digital converter (ADC) and one conditioning circuit. This is an advantage because the size of the circuitry is small, and the cost is also reduced. The multiplexer selected is HP4067 of Texas Instruments. This device allows digital control of 16 channels using CMOS technology and high switching speed [19]. The ESP32S controls the multiplexer. In order to read the 16 sensors, a sweep is performed with a frequency of 31 Hz. Finally, a Zener diode (1N4729A) is placed parallel to the voltage output to protect the ADC port of ESP32S. This diode limits the output voltage to 3.3 V.

3) *Power supply*: A lithium polymer (LiPo) battery supplies power to the system. The model selected has two cells and a capacity of 1000 mAh, which gives 7.4 V that in the circuit represents VSS. In order to get $V_{CC} = 5 \text{ V}$, a step-down voltage converter LM2596 was selected. It has a conduction capacity of 3 A. VCC is used to energize the conditioning circuit, EPS32S microcontroller, and the multiplexer.

4) *Data acquisition and Wi-Fi communication*: The ESP32S is in charge of acquiring data from the 16 sensors. It has a 12-bit resolution for its ADC. In this project, only one analog port is used for reading the output voltage of the conditioning circuit. The range of the converter is from 0-3.3 V. Moreover, five input/output digital ports are utilized for multiplexer control and to sweep the 16 sensors.

The ESP32S has a built-in Wi-Fi transmitter with a data speed transmission of 150 Mbps [20]. The communication protocol TCP/IP (Transmission control protocol/ Internet Protocol) and the model C/S (Client/ Server) are used. In order to create wireless communication, the ESP32S is configured as a server and the PC as a client. It is needed to establish a server port, and the client has to configure the same IP address of the server. For the user interface, LabView software is used. It is in charge of communicating with the microcontroller and then showing the data. LabView sends the instruction for reading a specific channel of the multiplexer. Then the microcontroller sends the voltage output of the conditioning circuit as a response. This process is repeated for the 16 channels of the multiplexer at a reading frequency of 31 Hz.

5) *User Interface*: Once LabView gets voltage data from the conditioning circuit, signal processing transforms voltage values into force values. In order to perform the transformation, the experimentally found linear model is used. The user interface compresses of (i) A 3D model in which, through a heat map, exerted forces are shown over the stump. As can be seen in Fig. 7, on the left side, there is the heat map of the 3D stump model, and on the right side, there is a sensor distribution over it. (ii) voltage and force curves vs. time are also displayed for each sensor. Fig. 8.a shows the voltage obtained from sensor 1 after a performed test and Fig. 8.b shows the force exerted over sensor 1. The complete interface presents 16 graphs of voltage and 16 of force due to each sensor.

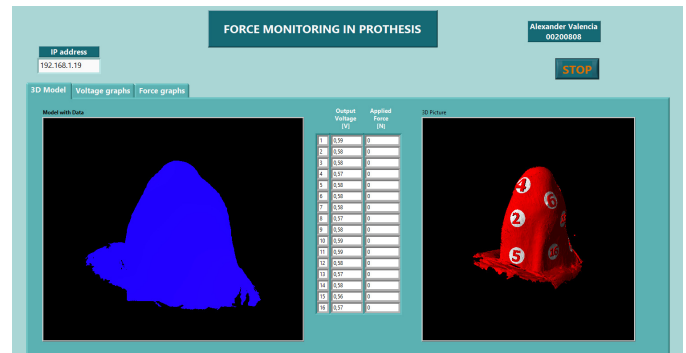


Fig. 7: Home page of the interface.

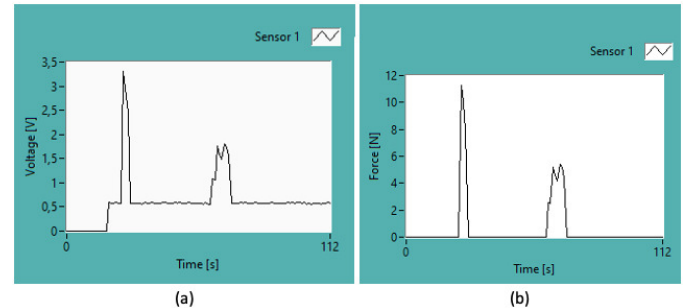


Fig. 8: Example of the sensor data shown in the interface: a) Voltage graph b) Force graph.

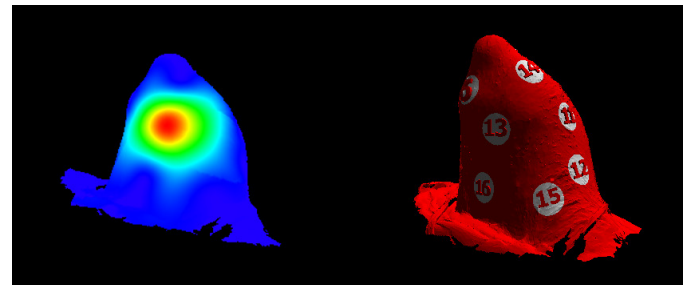


Fig. 9: 3D model - 10a) Heat map of one sensor activated, 10b) Sensor positions.

A 3D model of the stump is also developed and exported to LabView, in which the position of sensors are indicated together with the magnitude of applied force as a color scale as shown in Fig. 9.

6) *Printed circuit board (PCB) design*: One of the objectives of the prototype is to be portable. A compact design of the PCB was developed. Fig. 10 shows the components of the board: conditioning circuit, microcontroller, and multiplexer. The dimensions of this initial proof of concept version are 12.5 cm x 7 cm.

7) *Sensors over the stump*: For testing purposes, the sensors were distributed around the stump mold as shown in Fig. 11, so that the user interface displays the information of each sensor in real-time.

III. EXPERIMENTATION AND RESULTS

Once the conditioning circuit was developed, different tests with several sensors were performed to check the circuit. Forces between 1-11 N were applied, and the output voltages were measured. Table I shows data obtained. With these data,

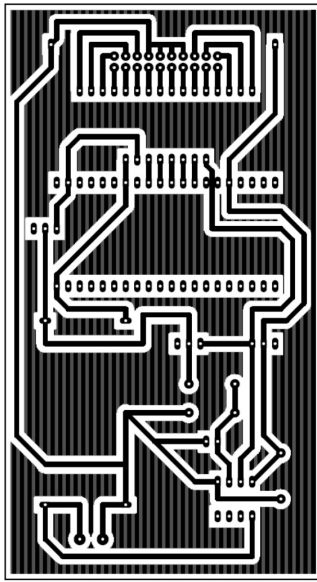


Fig. 10: PCB Design.

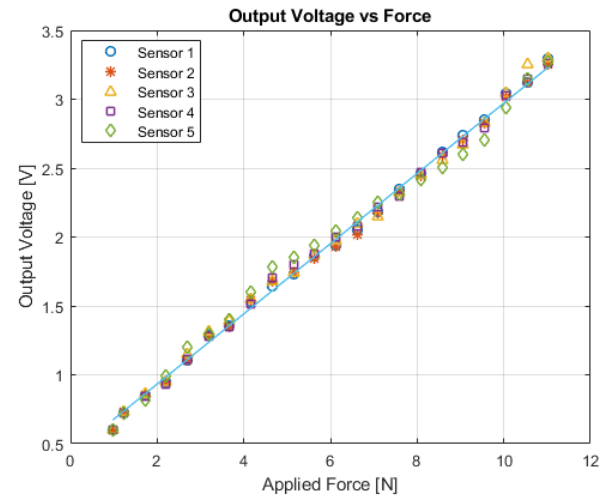


Fig. 12: Example of the sensor output: Voltage vs. Force graph.

TABLE I: Conditioning circuit voltage outputs

Force (N)	Sensor1 (V)	Sensor2 (V)	Sensor3 (V)	Sensor4 (V)	Sensor5 (V)
0.98	0.6	0.6	0.6	0.6	0.6
1.23	0.72	0.73	0.73	0.72	0.72
1.72	0.85	0.85	0.86	0.84	0.82
2.21	0.95	0.95	0.95	0.93	0.99
2.70	1.1	1.1	1.15	1.11	1.2
3.19	1.29	1.28	1.31	1.28	1.3
3.68	1.36	1.35	1.39	1.35	1.4
4.17	1.53	1.54	1.55	1.51	1.6
4.66	1.64	1.68	1.68	1.7	1.78
5.15	1.73	1.74	1.74	1.8	1.85
5.64	1.86	1.84	1.87	1.88	1.94
6.13	1.94	1.93	1.97	2	2.04
6.62	2.05	2.02	2.1	2.08	2.14
7.11	2.18	2.17	2.15	2.22	2.25
7.60	2.35	2.33	2.32	2.3	2.31
8.09	2.46	2.44	2.45	2.47	2.42
8.58	2.62	2.59	2.56	2.61	2.5
9.07	2.74	2.7	2.67	2.69	2.6
9.56	2.85	2.83	2.83	2.79	2.7
10.05	3.04	3.02	3.04	3.03	2.94
10.54	3.12	3.12	3.25	3.15	3.15
11.03	3.3	3.25	3.3	3.26	3.27

result when the stump model is inside the prosthesis (Fig. 14). As can be seen, on the left side, the magnitude of forces is presented as a heat map, where the colors red, orange, and yellow present the level of force in descending order, and the green color shows a lower force. The right side presents the position of each sensor. In between, a table with the information of voltage and force of each sensor is also presented.

Additionally, the system has two more windows with voltage graphs and forces vs. time. In this way, it is possible to determine where there is constant pressure and for how long, which represents a helpful tool to improve the design of the prosthesis in terms of comfort.

IV. CONCLUSIONS

The design and implementation of a real-time system for continuous force monitoring of a canine prosthesis were presented. A transimpedance amplifier was selected as a conditioning circuit because it gives a linear response between



Fig. 11: Sensor network distribution on the prosthesis.

a linear regression was performed as shown in Fig.12, and the values of (4) were obtained. Finally, this transformation is applied in LabView to show data in force units within this model.

$$F = 0.2548 * V + 0.4228 \quad (4)$$

A. Real-time monitoring

Real-time monitoring of applied forces is needed because it aids in checking that the sensor and system work properly. Moreover, monitoring forces must be done while a mechanic stimulus is performed over the stump. Fig. 13 displays the

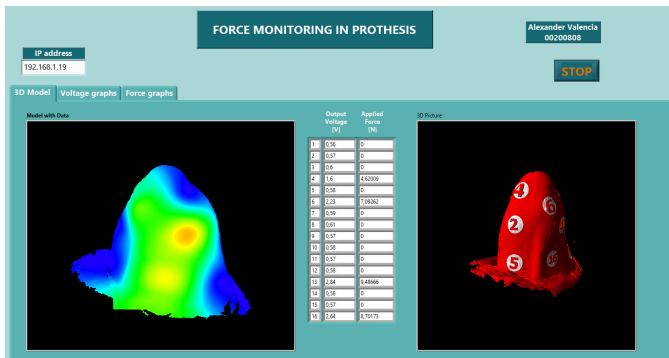


Fig. 13: Real-time monitoring system interface.



Fig. 14: Plaster cast of stump inside the prosthesis.

voltage and force exerted over sensors. A TCP/IP communication between a microcontroller and a PC was implemented so that the prototype is portable, working with a LiPo battery of two cells.

Initial results indicate that it is feasible to measure the forces applied to the stump of the canine with this monitoring system and analyze their distribution on the soft tissues of the residual limb. This is of great importance in optimizing the geometry of the custom-made socket to improve the comfort of the patient and prosthetic control. Additionally, an adequate fitting is important to decrease the positive pressure and avoid problems such as skin irritation, tissue breakdown, poor circulation, pain, and injuries.

The real-time monitoring system will help analyze pressure forces while the canine performs different activities, for example, analyzing the force distribution when the animal walks, runs, sits, or climbs stairs. The force vs. time graphs allow analyzing the data in real-time and over longer test periods. Together with motion capture technologies, kinematics, kinetics, and biomechanics can be better understood for an overall improvement in the functionality of the prosthesis.

As future work, we plan to use FSR sensors with a smaller active area to establish force points more accurately. Such results can help better understand the development of comfortable canine limb prosthesis designs that closely replicate the normal limb function.

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